

EXPERIMENT 21. Swap the Cases of a Specified Character Column

AIM

To write a Pandas program to swap the cases of a specified character column in a given DataFrame.

CODE

```
exp21.py > ...  
1  import pandas as pd  
2  
3  df = pd.DataFrame({  
4      'Name': ['Dharshini', 'Python', 'Pandas', 'MATPLOTLIB']  
5  })  
6  
7  df['Name'] = df['Name'].str.swapcase()  
8  
9  print(df)  
10
```

OUTPUT

```
Documents/Query process.  
      Name  
0  dHARSHINI  
1    pYTHON  
2    pANDAS  
3  matplotlib
```

RESULT

Thus, the cases of the specified character column were successfully swapped using Pandas.

EXPERIMENT 22. Draw a Line with Suitable Labels and Title

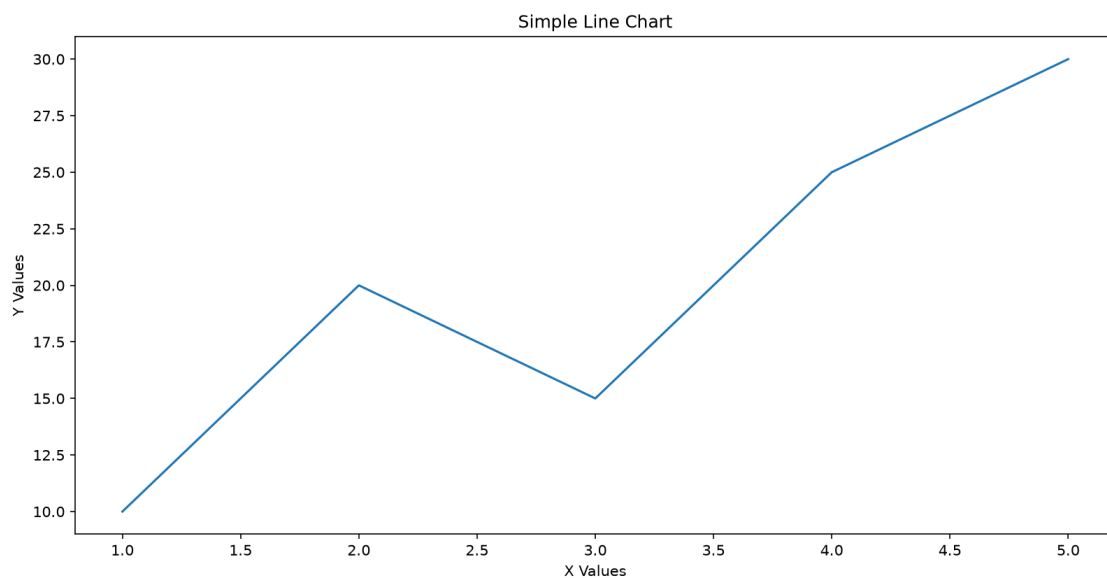
AIM

To write a Python program to draw a line chart with suitable labels for the X-axis, Y-axis, and title.

CODE

```
exp22.py > ...  
1  import matplotlib.pyplot as plt  
2  
3  x = [1, 2, 3, 4, 5]  
4  y = [10, 20, 15, 25, 30]  
5  
6  plt.plot(x, y)  
7  plt.xlabel("X Values")  
8  plt.ylabel("Y Values")  
9  plt.title("Simple Line Chart")  
10 plt.show()  
11
```

OUTPUT



RESULT

Thus, a line chart with suitable axis labels and title was successfully created.